



Analysis of Glucosylsphingosine in CSF samples from PPMI

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Summary

We have optimized a previously published, LC-MS/MS-based targeted assay (1) to measure Glucosylsphingosine (GlcSph) in human CSF, giving us the opportunity to accurately assess CSF GlcSph levels across both monogenic (GBA and LRRK2) and sporadic forms of Parkinson's Disease.

Method

To prepare samples for LC-MS/MS analysis, aliquots of 100 μ L of calibration standards (STDs), quality control samples (QCs), and experimental CSF samples were transferred to a clean 2-mL protein LoBind 96-well plate. 100 μ L of artificial CSF (aCSF) (Tocris Bioscience cat # 3525) was added to the STDs and QCs while 100 μ L of methanol was added to the CSF samples and CSF matrix QCs. 20 μ L of internal standard working solution (10 pg/mL of GlcSph-d5 (Avanti Polar Lipids cat # 860636P-1mg) in methanol) was added to each sample except blank samples while 20 μ L of methanol was added to the blank samples. 800 μ L of methanol/water/formic acid (450:550:5, v/v/v) was added to all samples. After mixing, the samples were loaded onto a 2.0 mL Strata-X Polymeric Reversed Phase plate (Phenomenex cat # 8E-S100-EGB), which was preconditioned with 1 mL of hexane twice followed by 1 mL of methanol twice and equilibrated with 1 mL of methanol/water/formic acid (450:550:5, v/v/v) twice. Once samples were completely absorbed in the plate sorbent, the plate was washed with 1 mL of methanol/water/formic acid (450:550:5, v/v/v) twice and 1 mL of methanol/water (450:550, v/v). After the washing, 1 mL protein LoBind 96-well collection plate was placed under the extraction plate. The analyte was eluted by 400 μ L of methanol/ammonium hydroxide (25%) (100:0.04, v/v) twice into a collection plate. The eluent was dried down completely under purified nitrogen gas flow at 60 °C and reconstituted in 120 μ L acetonitrile/methanol/water/formic acid/ammonium formate (1M) (2850:75:60:3:15, v/v/v/v/v) before the samples were injected to LC-MS/MS for analysis.

LC-MS/MS analysis were performed on an Exion LC AD UHPLC system coupled with a Sciex API 7500 Triple Quad mass spectrometer (AB Sciex, Redwood City, CA). The MRM transition





for the analyte and internal standard were 462.3 to 282.1 and 467.6 to 287.4, respectively. Collisional Energy was 27 v. HPLC chromatography was established on a Halo, 90 Å, HILIC column (2 µm, 3.0 x 150 mm) (Advance Materials Technology cat # 91813-701) and the column was kept at 35 °C during the run.

For the LC separation of GlcSph from the matrix interference, the two mobile phases used were 0.1 % Formic Acid and 50 mM ammonium formate in water (mobile phase A) and 0.1 % Formic Acid in acetonitrile/methanol (95:5, v/v) (mobile phase B). The flow rates were 0.5 mL/min for analyte elution and 0.3 mL/min for column wash and equilibration. Mobile phase B concentration was initially set at 91% and kept for 8.5 min, and then decreased to 20% at 8.6 min and kept until 10.60 min before increasing mobile phase B concentration to 91% at 10.70 min.

Data analysis

The GlcSph LC-MS/MS assay was performed in ten, equally distributed batches. These batches were divided into two rounds (4 batches in round 1 and 6 batches in round 2). All samples with remaining aliquots (n=568) were used from the original n=640 (see Project ID 180). A total of 2 pooled control and PD CSF sample replicates were created and distributed evenly within and between batches to enable evaluation of inter-batch deviations. Values below the quantification limit (BQL) were imputed by half of the lower limit of quantification (LLOQ) reported. The LLOQ varied based on volume of each CSF sample, thus, BQLs <0.2, <0.222, and <0.25 were imputed as 0.1, 0.111, and 0.125, respectively.

Data Availability

Analyte measurements from all study samples have been provided. Measured values in the pooled quality control samples are available upon request.

References:

- (1) Matsumoto SI, Sato S, Otake K, Kosugi Y. Highly-sensitive simultaneous quantitation of glucosylsphingosine and galactosylsphingosine in human cerebrospinal fluid by liquid chromatography/tandem mass spectrometry. J Pharm Biomed Anal. 2022 Aug 5;217:114852. doi: 10.1016/j.jpba.2022.114852. Epub 2022 May 23. PMID: 35636011.

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